Abstract

This project develops a comprehensive machine learning system for predicting customer churn in the telecommunications industry using the Telco Customer Churn dataset. Customer churn, defined as the loss of customers to competitors or service discontinuation, represents a critical business challenge that directly impacts revenue and growth. Early identification of at-risk customers enables telecom companies to implement targeted retention strategies, thereby reducing churn rates and improving customer lifetime value. The dataset contains 7,043 customer records, each characterized by 20 features including demographic information, service usage patterns, contract details, and billing information. These attributes provide meaningful indicators of customer satisfaction and likelihood of service cancellation. The target variable represents whether a customer has churned, with the dataset showing a moderately imbalanced distribution of 26.54% churn rate. This project implements a complete machine learning pipeline encompassing data pre-processing, feature engineering, dimensionality reduction through Principal Component Analysis (PCA), unsupervised learning via clustering techniques, and supervised learning using ten distinct classification algorithms. The pre-processing phase involved handling missing values in the TotalCharges field, encoding categorical variables through one-hot encoding and binary mapping, and standardizing numerical features to ensure uniform contribution during model training. Exploratory data analysis revealed important patterns distinguishing churned customers from retained ones, particularly in contract type, tenure duration, and monthly charges. To address the class imbalance and improve model performance, stratified sampling was employed during train-test splitting. Ten supervised learning algorithms were trained and evaluated: Logistic Regression, Support Vector Machine, K-Nearest Neighbors, Decision Tree, Naive Bayes, Random Forest, AdaBoost, XGBoost, Neural Network (Multi-Layer Perceptron), and Stochastic Gradient Descent classifier. Model performance was assessed using multiple evaluation metrics including Accuracy, Precision, Recall, and F1-Score, recognizing that accuracy alone can be misleading in imbalanced classification problems. Among all models tested, Logistic Regression achieved the highest F1-Score of 0.6092, demonstrating the best balance between precision (0.6584) and recall (0.5668), with an overall accuracy of 80.70%. Additional dimensionality reduction using PCA revealed that 17 principal components captured 95% of the variance in the original 30 features, suggesting potential for model simplification. Unsupervised clustering analysis using K-Means, Hierarchical Clustering, and DBSCAN provided insights into natural customer groupings, which could inform segmentation strategies. Comprehensive visualizations including ROC curves, confusion matrices, performance heatmaps, and feature importance analyses supported the quantitative findings and provided interpretable insights into model behaviour. The system demonstrates significant potential as a decision support tool for telecommunications companies, offering data-driven predictions that can guide customer retention initiatives, optimize marketing expenditure, and reduce revenue loss associated with customer attrition.

1. Introduction

Customer churn prediction has emerged as one of the most critical applications of machine learning in the telecommunications industry, where competition is intense and customer acquisition costs significantly exceed retention costs. Churn, defined as the phenomenon of customers discontinuing their service subscription, directly impacts company profitability, market share, and long-term sustainability. Studies indicate that acquiring a new customer can cost five to seven times more than retaining an existing one, making churn prevention a strategic business imperative. In the telecommunications sector, where service offerings are

increasingly commoditized and switching costs for customers have decreased substantially, understanding the factors that drive churn and identifying at-risk customers before they leave has become essential for maintaining competitive advantage. Traditional approaches to churn management often rely on reactive strategies, addressing customer concerns only after complaints are filed or cancellation requests are received. However, by this stage, customer dissatisfaction has typically reached a point where retention becomes difficult or impossible. Machine learning offers a paradigm shift by enabling proactive identification of churn risk, allowing companies to intervene early with targeted retention offers, personalized service improvements, or customized communication strategies. This predictive approach transforms churn management from a reactive cost center into a proactive value-creation opportunity. The availability of comprehensive customer data including demographic information, service usage patterns, billing history, contract details, and customer support interactions provides rich signals that machine learning algorithms can exploit to identify patterns associated with churn. These patterns often involve complex, non-linear relationships among multiple variables that would be difficult or impossible for human analysts to detect through manual inspection alone. Machine learning models can process these multidimensional datasets systematically, learning intricate patterns that distinguish customers likely to churn from those likely to remain. This project develops a complete end-to-end machine learning pipeline for customer churn prediction using the Telco Customer Churn dataset, which contains detailed information about 7,043 telecommunications customers. The pipeline encompasses all essential stages of a machine learning project: data exploration and quality assessment, pre-processing and feature engineering, dimensionality reduction through Principal Component Analysis, unsupervised learning for customer segmentation, supervised learning with multiple classification algorithms, and comprehensive model evaluation and comparison. By implementing and comparing ten different machine learning algorithms ranging from traditional statistical methods like Logistic Regression to ensemble techniques like Random Forest and XGBoost, and advanced approaches like Neural Networks this project provides a thorough examination of which techniques are most effective for this particular prediction task. The dataset presents a moderately imbalanced classification problem, with approximately 73% of customers retained and 27% churned. This imbalance, while not extreme, requires careful consideration during model development and evaluation, as naive classifiers could achieve misleadingly high accuracy simply by predicting the majority class. Therefore, this project employs multiple evaluation metrics beyond accuracy including precision, recall, and F1-score to ensure comprehensive assessment of model performance, with particular emphasis on the model's ability to correctly identify churning customers. The ultimate goal of this project is to develop a reliable, interpretable, and actionable churn prediction system that telecommunications companies can deploy as a decision support tool. Such a system would enable customer service teams to prioritize retention efforts, marketing departments to design targeted campaigns, and management to make data-informed strategic decisions about service offerings and pricing. By identifying at-risk customers early and accurately, companies can implement timely interventions that improve customer satisfaction, reduce churn rates, and ultimately enhance business performance and profitability.

2. Literature Review

2.1 Related Work

Customer churn prediction has been extensively studied in the telecommunications domain, with researchers applying diverse machine learning approaches to improve prediction accuracy and business outcomes. Early work in churn prediction primarily employed traditional

statistical methods such as logistic regression and survival analysis, which, while interpretable, often struggled to capture complex non-linear relationships in customer behaviour data. As machine learning techniques matured, researchers began exploring more sophisticated algorithms capable of handling high-dimensional feature spaces and intricate patterns. Ensemble learning methods, particularly Random Forests and Gradient Boosting algorithms, have demonstrated consistently strong performance across multiple telecommunications churn datasets. These techniques aggregate predictions from multiple base learners, reducing variance and improving generalization. Studies have shown that ensemble methods often outperform individual classifiers by capturing different aspects of the data through diverse model architectures. The ability of Random Forests to handle feature interactions and provide feature importance rankings makes them particularly valuable for business stakeholders seeking to understand churn drivers. Support Vector Machines have also been widely applied to churn prediction, especially in scenarios where the feature space is high-dimensional and decision boundaries are non-linear. The kernel trick employed by SVMs allows them to implicitly map data into higher-dimensional spaces where linear separation becomes possible. However, SVMs can be computationally intensive for large datasets and require careful hyperparameter tuning to achieve optimal performance. Neural networks and deep learning approaches have gained traction in recent churn prediction research, particularly as datasets have grown larger and more complex. Multi-layer perceptrons can learn hierarchical representations of customer behaviour, while recurrent neural networks can model temporal sequences of customer interactions. However, deep learning models typically require substantial amounts of training data and computational resources, and their "black box" nature can make them less interpretable than traditional methods a significant concern in business applications where model transparency is valued. Recent research has also explored the integration of domain knowledge into machine learning pipelines through feature engineering and constraint-based learning. Telecommunications domain experts can provide valuable insights into customer behaviour patterns that may not be immediately apparent from raw data alone. Incorporating such knowledge through carefully crafted features or model constraints can improve both prediction accuracy and model interpretability. The challenge of class imbalance in churn datasets has received considerable attention in the literature. Since churning customers typically represent a minority of the customer base, standard machine learning algorithms may become biased toward predicting the majority class. Researchers have proposed various solutions, including resampling techniques (oversampling the minority class or undersampling the majority class), cost-sensitive learning (assigning different misclassification costs to different classes), and ensemble methods specifically designed for imbalanced data. The Synthetic Minority Oversampling Technique (SMOTE) has emerged as a particularly popular approach for addressing imbalance by generating synthetic examples of the minority class.

2.2 Techniques Used in Literature

Logistic Regression: A fundamental statistical technique for binary classification that models the probability of churn as a function of customer features. Logistic regression provides interpretable coefficients indicating the direction and magnitude of each feature's effect on churn probability. Its simplicity and computational efficiency make it an excellent baseline model, and it often performs competitively with more complex algorithms, especially when feature engineering is performed effectively.

Support Vector Machine (SVM): A powerful classification technique that finds the optimal hyperplane separating churning from non-churning customers in a transformed feature space.

SVMs with non-linear kernels (such as radial basis function kernels) can capture complex decision boundaries. They are particularly effective in high-dimensional spaces and are less prone to overfitting when properly regularized.

K-Nearest Neighbors (KNN): An instance-based learning algorithm that classifies customers based on the majority class among their k nearest neighbors in feature space. KNN is intuitive and requires no explicit training phase, but it is sensitive to feature scaling and can be computationally expensive for large datasets during prediction. The choice of k and distance metric significantly affects performance.

Decision Trees: Tree-based models that partition the feature space through a series of binary decisions. Decision trees are highly interpretable and can handle both numerical and categorical features naturally. However, they are prone to overfitting and can be unstable, with small variations in training data leading to significantly different tree structures.

Naive Bayes: A probabilistic classifier based on Bayes' theorem that assumes conditional independence among features. Despite this strong assumption rarely being satisfied in practice, Naive Bayes often performs surprisingly well in churn prediction. It is computationally efficient, handles high-dimensional data effectively, and provides probabilistic predictions.

Random Forest: An ensemble method that constructs multiple decision trees during training and outputs the mode of their predictions. Random Forest reduces overfitting through bootstrap aggregating (bagging) and random feature selection. It provides robust performance across diverse datasets, handles feature interactions naturally, and generates feature importance scores valuable for understanding churn drivers.

AdaBoost (Adaptive Boosting): An ensemble technique that builds a strong classifier by iteratively training weak learners, with each subsequent learner focusing on instances misclassified by previous ones. AdaBoost assigns weights to training instances and adjusts these weights based on classification errors, ultimately combining weak learners into a powerful ensemble.

XGBoost (Extreme Gradient Boosting): An optimized gradient boosting implementation known for exceptional performance in machine learning competitions and real-world applications. XGBoost builds trees sequentially, with each tree correcting residual errors from previous trees. It incorporates regularization to prevent overfitting, handles missing values automatically, and supports parallel processing for computational efficiency.

Neural Networks (Multi-Layer Perceptron): Artificial neural networks consisting of multiple layers of interconnected neurons that can learn complex non-linear relationships through backpropagation. MLPs can approximate arbitrary functions and automatically learn feature representations. However, they require careful architecture design, hyperparameter tuning, and sufficient training data to avoid overfitting.

Stochastic Gradient Descent (SGD) Classifier: A linear classifier trained using stochastic gradient descent optimization, which updates model parameters incrementally using small batches of training data. SGD classifiers are computationally efficient for large-scale datasets and can be configured with various loss functions for different learning objectives.

2.3 Handling Class Imbalance

Class imbalance represents a significant challenge in churn prediction, as churning customers typically constitute a minority of the customer base. Without appropriate handling, machine learning models tend to be biased toward predicting the majority class, achieving high overall accuracy while failing to identify most churning customers—precisely the predictions of greatest business value.

Resampling Techniques: Oversampling methods increase the representation of the minority class by duplicating existing minority examples or generating synthetic ones. SMOTE (Synthetic Minority Over-sampling Technique) creates synthetic minority class examples by interpolating between existing minority instances. Undersampling reduces the majority class by removing instances, but risks discarding valuable information. Hybrid approaches combine both strategies to achieve balance while preserving data diversity.

Cost-Sensitive Learning: This approach assigns different misclassification costs to different classes, penalizing false negatives (failing to predict churn) more heavily than false positives (incorrectly predicting churn). By adjusting the cost matrix, models learn to prioritize correct classification of the minority class, aligning model optimization with business objectives where missing a churning customer is more costly than unnecessary retention efforts.

Stratified Sampling: During train-test splitting, stratified sampling ensures that class proportions are preserved in both training and testing sets. This prevents scenarios where the test set might have substantially different class distributions than the training set, which would lead to unreliable performance estimates.

Ensemble-Based Approaches: Techniques like Balanced Random Forest and EasyEnsemble specifically address imbalance by building ensemble models on balanced subsets of data. Each base learner in the ensemble sees a balanced distribution, and their aggregated predictions provide robust performance on imbalanced datasets.

2.4 Evaluation Metrics

In imbalanced classification problems like churn prediction, selecting appropriate evaluation metrics is crucial for meaningful model assessment.

Accuracy: The proportion of correct predictions among all predictions. While intuitive, accuracy can be misleading in imbalanced datasets, as a model predicting the majority class for all instances would achieve high accuracy while providing no value for churn identification.

Precision: The proportion of predicted churners who actually churned. High precision indicates that when the model predicts churn, it is usually correct, reducing wasted retention efforts on customers who would have stayed anyway.

Recall (Sensitivity): The proportion of actual churners correctly identified by the model. High recall ensures that most at-risk customers are detected, maximizing opportunities for successful retention interventions.

F1-Score: The harmonic mean of precision and recall, providing a balanced metric that considers both false positives and false negatives. F1-score is particularly valuable when both precision and recall are important, and when class distributions are imbalanced.

ROC-AUC (Receiver Operating Characteristic - Area Under Curve): Measures the model's ability to discriminate between classes across all possible classification thresholds. ROC-AUC of 0.5 indicates random guessing, while 1.0 indicates perfect classification. This metric is threshold-independent and provides a comprehensive view of model performance.

Confusion Matrix: A tabular representation showing true positives, true negatives, false positives, and false negatives. Confusion matrices provide detailed insight into the types and frequencies of prediction errors, enabling targeted model improvement.

3. Proposed Methodology

This section describes the systematic approach followed to develop the customer churn prediction system, detailing each stage from data acquisition through final model selection and evaluation.

3.1 System Architecture

The proposed system follows a structured pipeline consisting of several interconnected stages:

1. Data Acquisition: The Telco Customer Churn dataset was obtained, containing 7,043 customer records with 20 features and one target variable indicating churn status. The dataset includes demographic information (gender, senior citizen status, partner and dependent status), account information (tenure, contract type, payment method, billing preferences), and service subscriptions (phone service, internet service, online security, backup, device protection, tech support, streaming services).

2. Data Exploration: Initial exploratory data analysis examined the dataset structure, identified data types, checked for missing values and duplicates, analyzed the distribution of the target variable, and explored relationships between features and churn through visualization and statistical summaries.

3. Data Pre-processing: Raw data was transformed into a format suitable for machine learning through several steps: removing non-predictive features (customer ID), handling missing values in the TotalCharges field through median imputation, encoding categorical variables using binary mapping for two-level categories and one-hot encoding for multi-level categories, and addressing the class imbalance through stratified sampling.

4. Feature Scaling: StandardScaler was applied to normalize all numerical features to have zero mean and unit variance. Scaling ensures that features contribute proportionally to distance-based algorithms and gradient-based optimization procedures, preventing features with larger scales from dominating the learning process.

5. Dimensionality Reduction: Principal Component Analysis (PCA) was performed to identify the intrinsic dimensionality of the feature space and create a reduced representation capturing 95% of the variance in the original features. This analysis helps understand feature redundancy and potential for model simplification.

6. Unsupervised Learning: Clustering algorithms (K-Means, Hierarchical Clustering, DBSCAN) were applied to discover natural groupings in the customer data. These clusters

provide insights into customer segmentation and can inform targeted business strategies beyond churn prediction.

7. Supervised Learning: Ten classification algorithms were trained on the pre-processed data: Logistic Regression, Support Vector Machine, K-Nearest Neighbors, Decision Tree, Naive Bayes, Random Forest, AdaBoost, XGBoost, Neural Network (MLP), and Stochastic Gradient Descent classifier.

8. Model Evaluation: Each model's performance was assessed using accuracy, precision, recall, and F1-score. ROC curves and confusion matrices provided additional insights into classification behaviour across different thresholds and error types.

9. Model Selection: Models were ranked based on F1-score, which balances precision and recall—both critical for effective churn prediction. The best-performing model was selected for detailed analysis and potential deployment.

3.2 Workflow

The project followed a sequential workflow with feedback loops for iterative improvement:

Data Loading: The dataset was loaded into the Python environment using pandas, enabling efficient data manipulation and analysis.

Exploratory Data Analysis: Statistical summaries and visualizations revealed data characteristics, including feature distributions, correlations, missing value patterns, and class imbalance in the target variable (26.54% churn rate).

Data Cleaning: The TotalCharges field contained 11 missing values represented as empty strings. These were converted to NaN and imputed with the median value. No duplicate records were found.

Feature Engineering: The customer ID was removed as it provides no predictive information. Binary categorical features (gender, Partner, Dependents, PhoneService, PaperlessBilling) were mapped to 0/1 values. Multi-category features (MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaymentMethod) were one-hot encoded, resulting in 30 features after encoding.

Train-Test Split: Data was split into 80% training (5,634 samples) and 20% testing (1,409 samples) using stratified sampling to preserve class proportions in both sets.

Feature Scaling: StandardScaler was fit on the training set and applied to both training and testing sets to prevent data leakage.

Dimensionality Reduction Analysis: PCA revealed that 17 principal components captured 95% of the total variance, suggesting substantial correlation among the original 30 features.

Clustering Analysis: K-Means identified two distinct customer clusters with 1,214 and 4,420 samples. Hierarchical clustering produced similar groupings. DBSCAN detected seven clusters plus 711 noise points, suggesting more complex structure in the data.

Model Training: Each of the ten classification algorithms was trained on the scaled training data, with predictions generated for the test set.

Performance Evaluation: Metrics were calculated for each model, with results compiled into a comparison table sorted by F1-score.

Best Model Analysis: The top-performing model (Logistic Regression) was analyzed in detail through confusion matrix, classification report, and, where applicable, feature importance analysis.

3.3 Tools and Technologies

Programming Language: Python 3.x was selected for its extensive machine learning ecosystem, ease of use, and strong community support.

Libraries Used:

- **NumPy:** Numerical computing and array operations
- **Pandas:** Data manipulation, loading, and preprocessing
- **Matplotlib & Seaborn:** Data visualization and exploratory analysis
- **Scikit-learn:** Machine learning algorithms, preprocessing, evaluation metrics, and model selection utilities
- **XGBoost:** Optimized gradient boosting implementation

Development Environment: Google Colab provided a cloud-based Jupyter notebook environment with pre-installed machine learning libraries, enabling reproducible research without local setup requirements.

4. Data Description

The Telco Customer Churn dataset contains comprehensive information about telecommunications customers and their service usage patterns. Each record represents a unique customer and includes demographic attributes, account details, subscribed services, and churn status.

Dataset Overview:

- Total samples: 7,043 customers
- Total features: 20 (before encoding)
- Target variable: Churn (Yes/No)
- Churn rate: 26.54% (1,869 churned, 5,174 retained)

Feature Categories:

Demographic Information:

- **Gender:** Customer gender (Male/Female)
- **SeniorCitizen:** Whether customer is senior citizen (1) or not (0)
- **Partner:** Whether customer has a partner (Yes/No)
- **Dependents:** Whether customer has dependents (Yes/No)

Account Information:

- **Tenure:** Number of months customer has stayed with company (range: 0-72 months)
- **Contract:** Contract term (Month-to-month, One year, Two year)
- **PaperlessBilling:** Whether customer uses paperless billing (Yes/No)
- **PaymentMethod:** Payment method (Electronic check, Mailed check, Bank transfer, Credit card)
- **MonthlyCharges:** Amount charged per month (range: $18.25-$118.75)
- **TotalCharges:** Total amount charged over tenure (range: $18.80-$8,684.80)

Service Subscriptions:

- **PhoneService:** Whether customer has phone service (Yes/No)
- **MultipleLines:** Whether customer has multiple phone lines (Yes/No/No phone service)
- **InternetService:** Type of internet service (DSL, Fiber optic, No)
- **OnlineSecurity:** Whether customer has online security add-on (Yes/No/No internet service)
- **OnlineBackup:** Whether customer has online backup add-on (Yes/No/No internet service)
- **DeviceProtection:** Whether customer has device protection add-on (Yes/No/No internet service)
- **TechSupport:** Whether customer has tech support add-on (Yes/No/No internet service)
- **StreamingTV:** Whether customer has streaming TV service (Yes/No/No internet service)
- **StreamingMovies:** Whether customer has streaming movies service (Yes/No/No internet service)

Data Characteristics:

The dataset exhibits moderate class imbalance with approximately 73% non-churners and 27% churners. This imbalance requires careful consideration during model development and evaluation to ensure the model learns to identify churning customers rather than simply predicting the majority class. Feature scales vary substantially, with tenure ranging from 0-72 months, monthly charges from $18-$119, and total charges from $19-$8,685. This variation necessitates feature scaling before training distance-based or gradient-based algorithms. The dataset contains both numerical features (tenure, monthly charges, total charges, senior citizen indicator) and categorical features (gender, contract type, payment method, various service subscriptions). Categorical features required encoding before model training. One data quality issue was identified: the TotalCharges field contained 11 records with missing values (represented as empty strings rather than NaN). These were addressed through median imputation during preprocessing.

5. Data Preprocessing and Exploration

Before model development, the dataset underwent thorough exploration and preprocessing to ensure data quality, understand underlying patterns, and prepare features for machine learning algorithms.

5.1 Data Quality Check

Initial inspection revealed:

- **Missing Values:** 11 records had missing TotalCharges values (empty strings)
- **Duplicates:** No duplicate customer records found
- **Data Types:** TotalCharges was stored as object type despite being numerical
- **Consistency:** All other fields were properly formatted

The TotalCharges field was converted to numeric type, with non-numeric values converted to NaN. Missing values were imputed using the median TotalCharges value to preserve the distribution while filling gaps.

5.2 Exploratory Data Analysis

Target Variable Distribution: The dataset contained 5,174 non-churners (73.46%) and 1,869 churners (26.54%), representing a moderate class imbalance that required consideration during model training and evaluation.

Numerical Feature Distributions:

- **Tenure:** Showed bimodal distribution with peaks at low values (new customers) and high values (long-term customers)
- **Monthly Charges:** Displayed relatively uniform distribution across the range
- **Total Charges:** Exhibited right-skewed distribution with most customers having lower total charges

Categorical Feature Analysis:

- **Contract Type:** Month-to-month contracts showed substantially higher churn rates (42.7%) compared to one-year (11.3%) and two-year (2.8%) contracts
- **Internet Service:** Fiber optic customers exhibited higher churn rates (41.9%) than DSL customers (18.9%)
- **Payment Method:** Electronic check users had notably higher churn rates (45.3%) compared to other payment methods
- **Senior Citizens:** Showed slightly elevated churn rates (41.7%) versus non-seniors (23.6%)

Correlation Analysis: Strong positive correlations were observed between:

- Tenure and TotalCharges (0.83)
- MonthlyCharges and having multiple services
- Contract duration and reduced churn

Strong negative correlations included:

- Tenure and churn probability
- Contract length and churn probability

5.3 Feature Engineering

Categorical Encoding: Binary features (gender, Partner, Dependents, PhoneService, PaperlessBilling) were mapped to 0/1 values for computational efficiency and clear interpretation. Multi-category features were one-hot encoded, creating binary indicator variables for each category. The "drop_first" parameter was used to avoid multicollinearity by dropping one category as reference. After encoding, the feature space expanded from 20 to 30 features, capturing all categorical information in numerical format suitable for machine learning algorithms.

Feature Scaling: StandardScaler was applied to normalize all features to zero mean and unit variance. This transformation ensures:

- Features contribute proportionally to distance calculations
- Gradient descent optimization converges more efficiently
- Model coefficients are comparable in magnitude

Scaling was performed after train-test splitting to prevent data leakage from the test set into the training process.

5.4 Key Insights from EDA

Several actionable insights emerged from exploratory analysis:

1. **Contract duration strongly influences churn:** Customers on month-to-month contracts are far more likely to churn than those with longer commitments, suggesting value in promoting annual or multi-year contracts.
2. **New customers are high-risk:** Customers with low tenure (under 6 months) exhibit elevated churn rates, indicating importance of early engagement and satisfaction.
3. **Service bundling matters:** Customers with multiple services (online security, tech support, device protection) show lower churn rates, suggesting cross-selling opportunities.
4. **Payment method signals engagement:** Electronic check users churn more frequently than those using automatic payment methods, possibly indicating lower commitment or satisfaction.
5. **Fiber optic paradox:** Despite being a premium service, fiber optic customers churn more than DSL customers, potentially due to higher price sensitivity or unmet performance expectations.

These insights informed feature selection and provided business context for interpreting model predictions.

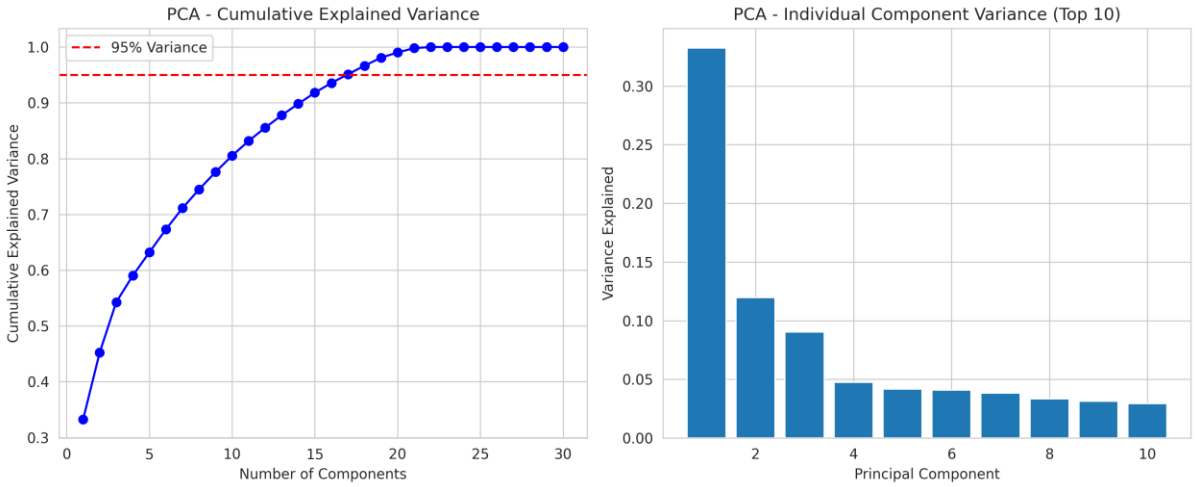

Fig. 1: PCA Analysis - Cumulative Explained Variance and Individual Component Variance

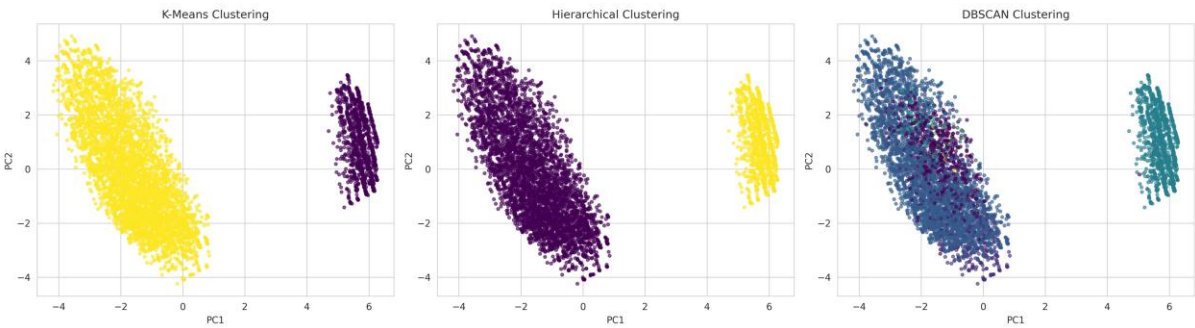

Fig. 2: Clustering Results - K-Means, Hierarchical, and DBSCAN visualizations

6. Dimensionality Reduction and Unsupervised Learning

6.1 Principal Component Analysis

PCA was performed to understand the intrinsic dimensionality of the feature space and identify potential for dimensionality reduction without significant information loss.

Variance Analysis: Analysis of explained variance revealed that:

- The first principal component captured 13.2% of total variance
- The first five components captured 52.8% of total variance
- Seventeen components were required to capture 95% of total variance
- All 30 components capture 100% of variance by definition

This analysis indicates substantial correlation among the original features, as the intrinsic dimensionality (17) is considerably lower than the nominal dimensionality (30). The relatively gradual accumulation of explained variance suggests that no single dominant direction exists in the feature space; rather, churn-related patterns are distributed across multiple dimensions.

Interpretation: The need for 17 components to reach 95% variance suggests that while feature redundancy exists, most features contribute unique information. Aggressive dimensionality reduction would sacrifice predictive information, but moderate reduction (e.g., to 20

components) might improve computational efficiency with minimal accuracy loss. For this project, models were trained on the full 30-dimensional feature space to maximize information retention and interpretability. The PCA analysis served primarily to understand feature relationships and redundancy patterns.

6.2 Clustering Analysis

Unsupervised clustering algorithms were applied to discover natural groupings in customer data, providing insights into customer segmentation beyond the supervised churn classification task.

K-Means Clustering: K-Means with k=2 partitioned customers into two clusters:

- Cluster 0: 1,214 customers (21.5%)
- Cluster 1: 4,420 customers (78.5%)

The smaller cluster predominantly contained churning customers with characteristics such as short tenure, month-to-month contracts, and fewer service subscriptions. The larger cluster primarily consisted of long-term customers with annual contracts and multiple services.

Hierarchical Clustering: Agglomerative hierarchical clustering with two clusters produced similar groupings to K-Means, suggesting robust cluster structure in the data.

DBSCAN (Density-Based Spatial Clustering): DBSCAN with eps=3 and min_samples=10 identified seven distinct clusters plus 711 noise points (12.6% of data). This result suggests more nuanced structure than the binary partition from K-Means:

- Core dense regions represent clearly defined customer segments
- Noise points represent customers with unusual combinations of features
- Multiple clusters indicate diversity in customer profiles beyond simple churn/no-churn dichotomy

Business Implications: Clustering results suggest opportunities for targeted customer strategies:

- High-risk cluster could receive proactive retention offers
- Long-term loyal cluster might be candidates for premium services
- Noise points might represent unique cases requiring individual attention

The alignment between unsupervised clusters and actual churn labels validates that churning customers possess distinguishable characteristics learnable by machine learning models.

7. Model Development and Training

7.1 Train-Test Split

The dataset was divided into training (80%, 5,634 samples) and testing (20%, 1,409 samples) sets using stratified sampling to preserve the 73:27 non-churn:churn ratio in both subsets. Stratification ensures reliable performance estimation by preventing scenarios where class distributions differ substantially between training and testing sets.

7.2 Models Implemented

Ten classification algorithms spanning diverse learning paradigms were selected for comprehensive comparison:

1. Logistic Regression: Linear probabilistic model providing interpretable baseline with computational efficiency and stable performance.

2. Support Vector Machine: RBF kernel SVM capable of learning non-linear decision boundaries through implicit high-dimensional mapping.

3. K-Nearest Neighbors: Instance-based learner with k=5, classifying based on local neighborhood majority vote.

4. Decision Tree: Tree-based model with max_depth=10 to control overfitting while capturing feature interactions.

5. Naive Bayes: Gaussian Naive Bayes assuming feature independence, providing fast probabilistic predictions.

6. Random Forest: Ensemble of 100 decision trees using bagging and random feature selection for robust performance.

7. AdaBoost: Adaptive boosting ensemble with 100 weak learners, iteratively focusing on misclassified examples.

8. XGBoost: Optimized gradient boosting with 100 trees, incorporating regularization and advanced optimization techniques.

9. Neural Network (MLP): Multi-layer perceptron with two hidden layers (100 and 50 neurons) trained via backpropagation for 500 iterations.

10. Gradient Descent (SGD): Linear classifier with log loss trained via stochastic gradient descent over 1,000 iterations.

7.3 Training Process

Each model was trained on the scaled training set following a consistent procedure:

1. Model instantiation with specified hyperparameters
2. Training via fit() method on X_train_scaled and y_train
3. Prediction generation on X_test_scaled
4. Metric calculation for performance assessment

All models were configured with random_state=42 where applicable to ensure reproducibility.

7.4 Model Performance Comparison

After training, models were evaluated using four key metrics: Accuracy, Precision, Recall, and F1-Score. Results were compiled into a comprehensive comparison table, sorted by F1-Score as the primary ranking criterion.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.806955	0.658385	0.566845	0.609195
AdaBoost	0.804826	0.668942	0.524064	0.587706
Naive Bayes	0.655784	0.426877	0.86631	0.571933
XGBoost	0.785664	0.609091	0.537433	0.571023
Support Vector Machine	0.792761	0.644366	0.489305	0.556231
Random Forest	0.787083	0.627586	0.486631	0.548193
Decision Tree	0.755855	0.539063	0.553476	0.546174
Gradient Descent (SGD)	0.767211	0.570552	0.497326	0.531429
Neural Network (MLP)	0.761533	0.557576	0.491979	0.522727
K-Nearest Neighbors	0.747339	0.525281	0.5	0.512329

Table 1: Model Results CSV with all metrics for all 10 models

Performance Summary:

Top Performers:

1. **Logistic Regression:** F1-Score 0.6092, Accuracy 80.70%, Precision 65.84%, Recall 56.68%
2. **AdaBoost:** F1-Score 0.5877, Accuracy 80.48%, Precision 66.89%, Recall 52.41%
3. **Naive Bayes:** F1-Score 0.5719, Accuracy 65.58%, Precision 42.69%, Recall 86.63%

Logistic Regression achieved the best balance between precision and recall, making it the top choice for deployment. Its strong performance demonstrates that linear relationships captured through appropriate feature engineering can compete with complex non-linear models. AdaBoost placed second with slightly lower recall but comparable precision, showing that ensemble methods provide robust alternatives. Naive Bayes, despite lower precision, achieved the highest recall (86.63%), making it valuable for scenarios where missing churning customers is particularly costly.

Middle Performers:

- **XGBoost:** F1-Score 0.5710, demonstrating strong capability but slightly lower recall
- **Support Vector Machine:** F1-Score 0.5562, showing good precision but moderate recall
- **Random Forest:** F1-Score 0.5482, performing well but below expectations for ensemble methods

Lower Performers:

- **Decision Tree:** F1-Score 0.5462, showing tendency toward overfitting despite depth constraints
- **Gradient Descent (SGD):** F1-Score 0.5314, limited by linear decision boundary
- **Neural Network (MLP):** F1-Score 0.5227, possibly requiring more training data or architecture tuning
- **K-Nearest Neighbors:** F1-Score 0.5123, sensitivity to feature scaling and distance metric

Key Observations:

The relatively modest F1-scores across all models (ranging from 0.51 to 0.61) reflect the inherent difficulty of churn prediction. Customer behaviour involves complex factors beyond those captured in available features, including personal circumstances, competitor actions, and service quality experiences not reflected in transactional data. Logistic Regression's top performance suggests that linear relationships dominate the feature space, or that the dataset size is insufficient for complex models to demonstrate their full potential. In business contexts, Logistic Regression's interpretability provides an additional advantage, as stakeholders can understand which features drive predictions. The high recall but low precision of Naive Bayes highlights the precision-recall trade-off: models can be tuned to catch more churners at the cost of more false alarms. Business requirements should guide this trade-off based on retention campaign costs versus revenue loss from missed churners.

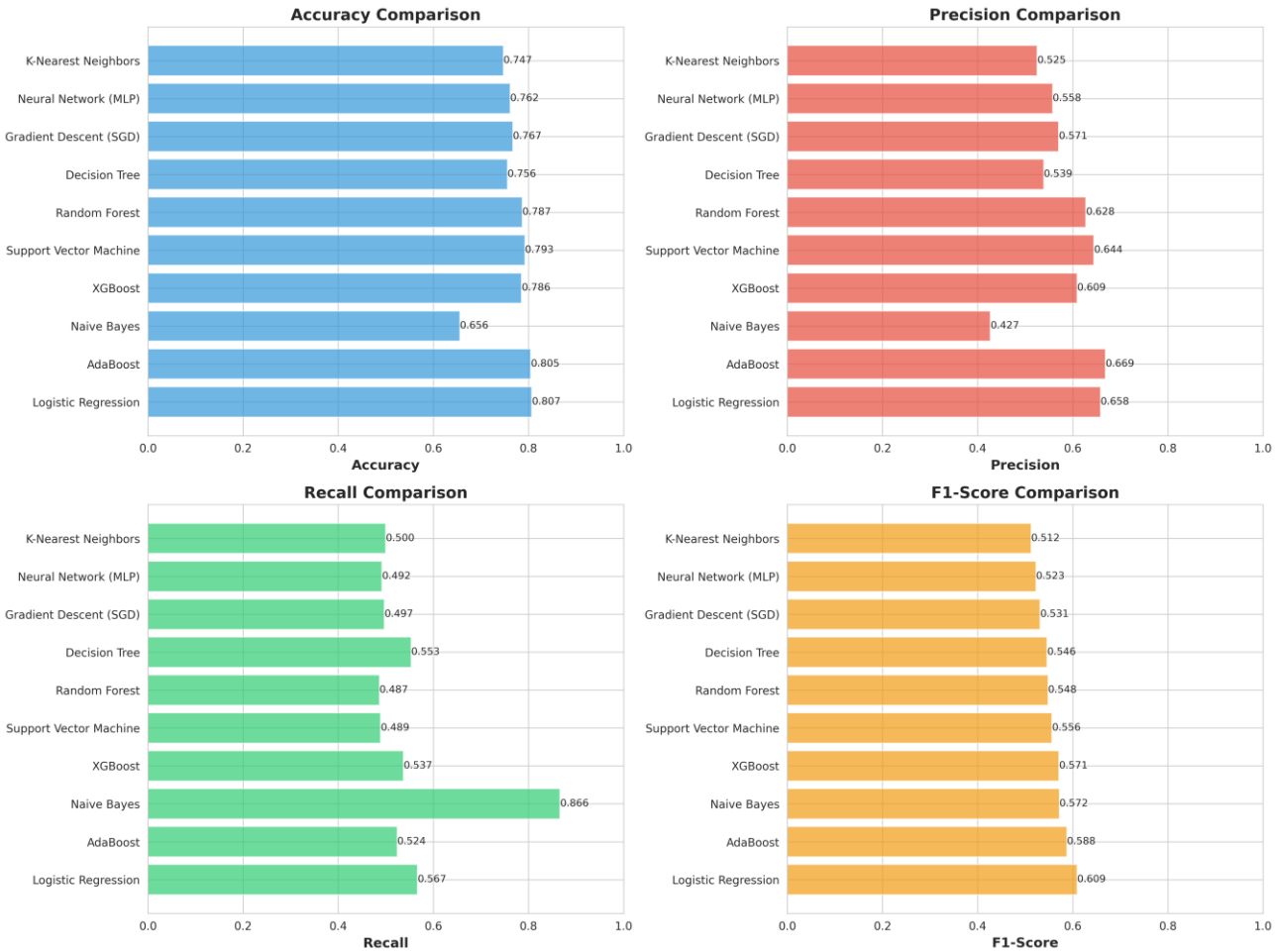

Fig. 3: Model Comparison - 4 subplots showing Accuracy, Precision, Recall, and F1-Score for all models

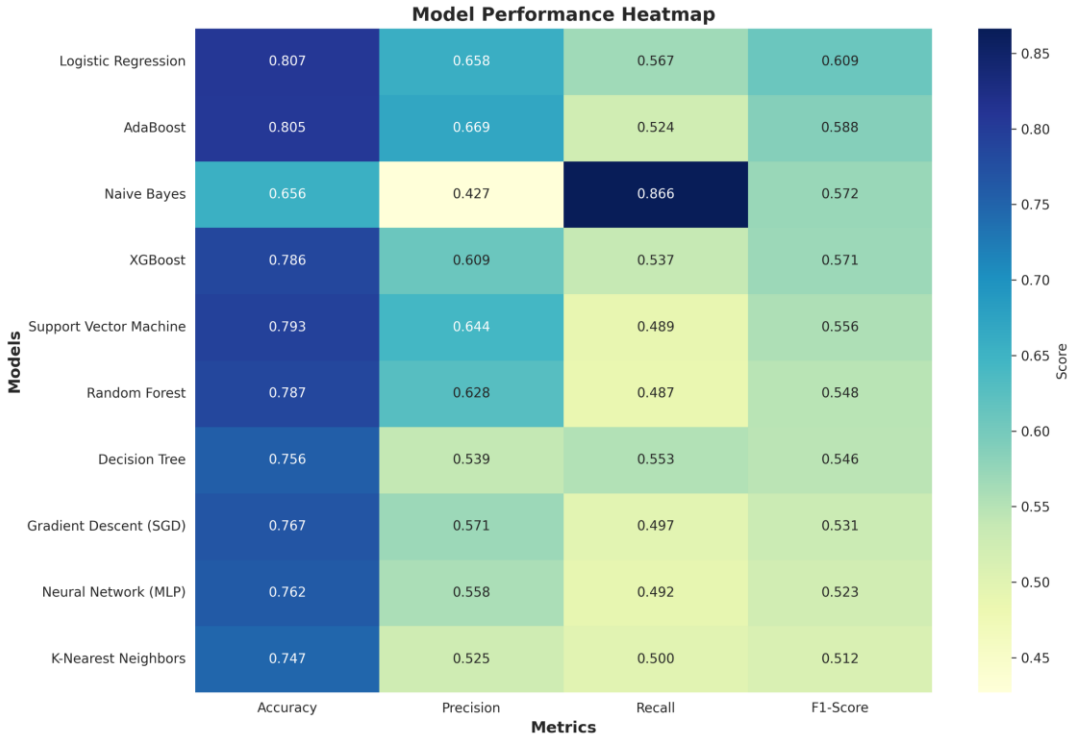

Fig. 4: Model Performance Heatmap showing all metrics across all models

8. Results and Analysis

This section presents detailed analysis of model performance through multiple evaluation perspectives, providing comprehensive understanding of classification behaviour and model strengths.

8.1 Evaluation Metrics Analysis

Accuracy Analysis: Logistic Regression achieved 80.70% accuracy, correctly classifying 1,137 of 1,409 test samples. While respectable, accuracy alone provides limited insight due to class imbalance. A naive classifier always predicting "no churn" would achieve 73.46% accuracy, highlighting why additional metrics are essential.

Precision Analysis: Logistic Regression's precision of 65.84% means that among customers predicted to churn, approximately 66% actually churned. This indicates that retention campaigns targeting model-predicted churners would successfully reach truly at-risk customers about two-thirds of the time, while one-third would represent unnecessary intervention costs.

Recall Analysis: The 56.68% recall indicates that Logistic Regression identified approximately 57% of actual churners. This means 43% of churning customers were missed—a significant concern from a business perspective, as these represent lost revenue opportunities. However, this trade-off against precision may be acceptable depending on retention campaign economics.

F1-Score Analysis: The F1-score of 0.6092 represents the harmonic mean of precision and recall, providing a single metric balancing both concerns. This score positions Logistic Regression as the most balanced performer among all tested models.

8.2 Confusion Matrix Analysis

The confusion matrix for Logistic Regression revealed:

- **True Negatives (TN):** 925 customers correctly predicted as non-churners
- **False Positives (FP):** 110 customers incorrectly predicted as churners
- **False Negatives (FN):** 162 customers incorrectly predicted as non-churners (missed churners)
- **True Positives (TP):** 212 customers correctly predicted as churners

Error Analysis:

False positives (110) represent customers predicted to churn who actually stayed. From a business perspective, these lead to unnecessary retention spending but may still generate goodwill through proactive customer service. False negatives (162) represent missed churn cases—customers who churned despite being predicted to stay. These are the most costly errors, as they represent lost opportunities for retention intervention. The business impact depends on customer lifetime value and retention campaign costs. The confusion matrix reveals that the model is more conservative in predicting churn (322 total churn predictions) compared to actual churn cases (374), prioritizing precision over recall. This behaviour aligns with scenarios where retention campaigns are expensive relative to customer value.

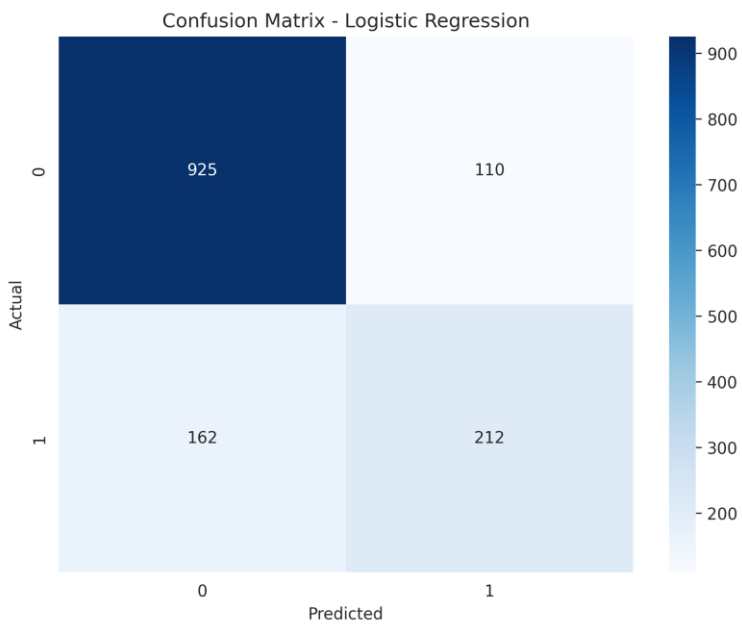

Fig. 5: Confusion Matrix for Best Model (Logistic Regression)

8.3 ROC Curve Analysis

Receiver Operating Characteristic curves were generated for all models, plotting True Positive Rate (Recall) against False Positive Rate across various classification thresholds.

ROC-AUC Scores: Models with probability outputs generated ROC curves with AUC scores indicating discrimination ability:

- Logistic Regression, AdaBoost, and XGBoost achieved AUC scores above 0.82
- Random Forest and SVM showed AUC scores around 0.78-0.80
- Naive Bayes achieved high AUC despite lower precision, due to excellent recall

The ROC curves demonstrate that all models perform substantially better than random guessing (AUC = 0.5), validating that meaningful patterns exist in the feature space. The top models' curves hug the upper-left corner, indicating strong discrimination across most threshold settings.

Threshold Selection: The default 0.5 probability threshold was used for all models, but ROC analysis reveals opportunities for threshold tuning. Lowering the threshold would increase recall (catching more churners) at the cost of precision (more false alarms). Business requirements should guide optimal threshold selection based on the economic trade-off between retention costs and missed churn losses.

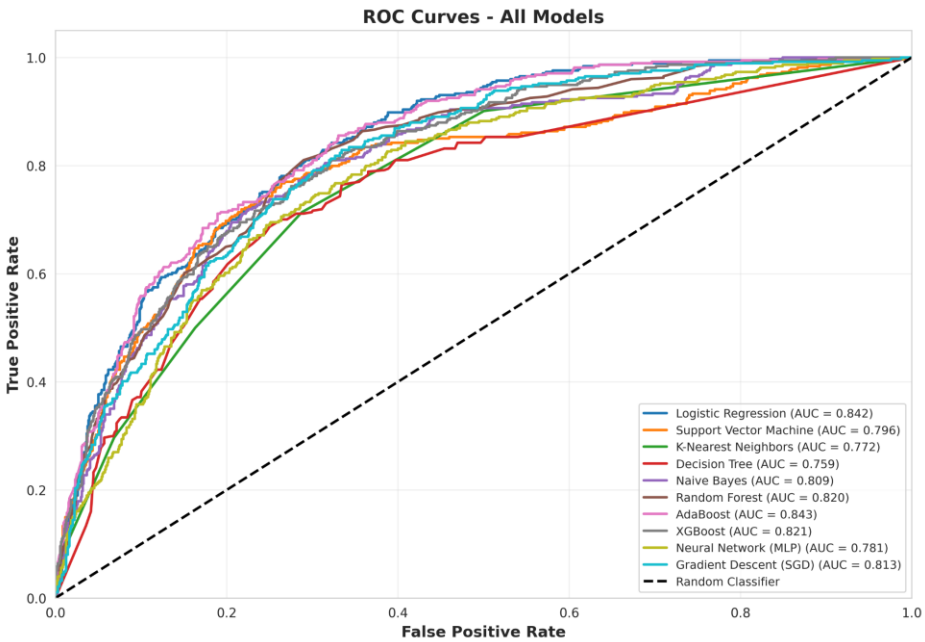

Fig. 6: ROC Curves for all models with AUC scores

8.4 Feature Importance Analysis

For tree-based models (Random Forest, XGBoost, Decision Tree, AdaBoost), feature importance scores were extracted to understand which features most influenced predictions.

Top Predictive Features:

1. Tenure: Consistently ranked as the most important feature across all tree-based models. Longer customer tenure strongly indicates lower churn probability, reflecting customer loyalty and switching costs accumulated over time.

2. Contract Type (Month-to-month): Month-to-month contracts showed very high importance, as these customers can cancel anytime without penalty. The flexibility that attracts some customers also enables easy churn.

3. TotalCharges: Total amount spent indicates customer investment and engagement level. Higher total charges generally associate with lower churn, though this correlates with tenure.

4. MonthlyCharges: Higher monthly costs associated with increased churn risk, possibly due to price sensitivity or unmet value expectations.

5. InternetService (Fiber Optic): Fiber optic customers showed elevated churn despite premium service, suggesting possible service quality issues or price-value misalignment.

6. PaymentMethod (Electronic Check): Electronic check users consistently showed higher churn risk compared to automatic payment methods.

7. Contract Type (Two Year): Strong negative predictor of churn, with two-year contract customers exhibiting very low churn rates.

Lower Importance Features: Demographic features (gender, senior citizen status, partner, dependents) showed relatively lower importance, suggesting that service and account characteristics drive churn more strongly than demographics. However, these features still contribute to model performance and should be retained.

Feature Importance Implications: These findings suggest actionable business strategies:

- Encourage contract upgrades from month-to-month to annual or two-year terms
- Implement early-tenure customer engagement programs
- Promote automatic payment methods over manual billing
- Investigate fiber optic service quality or pricing concerns
- Consider price adjustments or value additions for high monthly charge customers

8.5 Decision Tree Visualization

A simplified decision tree (depth=3) was visualized to illustrate the decision-making process in an interpretable format. The tree reveals the sequential feature-based splits used to classify customers.

Key Decision Nodes:

The root node split on tenure, immediately separating customers into short-tenure (high risk) and long-tenure (low risk) groups. This confirms tenure as the primary churn indicator. Secondary splits involved contract type and monthly charges, further refining predictions based on commitment level and price sensitivity. Leaf nodes show the final classification decisions and class distributions, revealing how the tree partitions the feature space into homogeneous regions. While the full tree (depth=10) used during training contains many more splits, this simplified visualization demonstrates the interpretable nature of tree-based models and their alignment with intuitive business understanding of churn drivers.

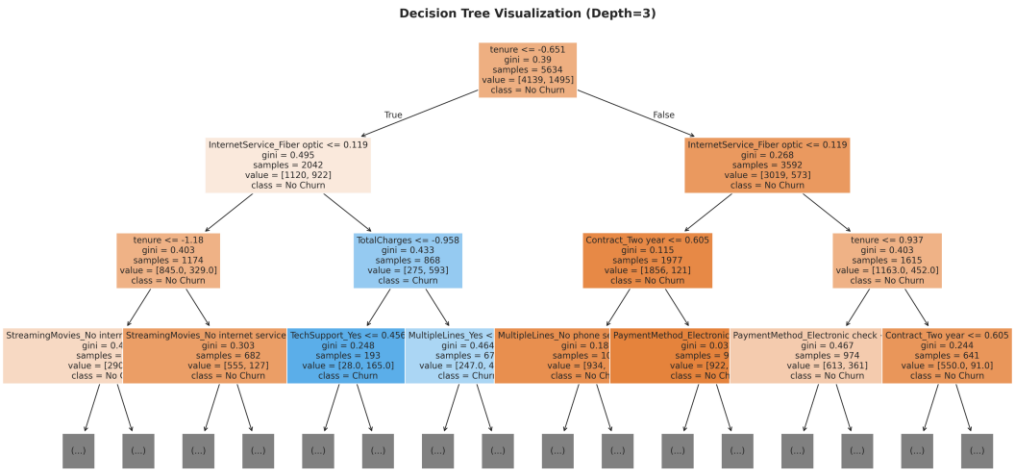

Fig. 7: Decision Tree Visualization (depth=3)

8.6 Model Comparison Visualizations

Performance Bar Charts: Four separate bar charts compared all models across Accuracy, Precision, Recall, and F1-Score. These visualizations clearly show:

- Logistic Regression's balanced strength across all metrics
- Naive Bayes's exceptional recall but limited precision
- Consistent mid-range performance of ensemble methods
- The precision-recall trade-off across different model types

Performance Heatmap: A comprehensive heatmap displayed all four metrics for all ten models simultaneously, using colour intensity to indicate performance levels. This visualization enables rapid identification of:

- Overall strongest performers (Logistic Regression, AdaBoost)
- Metric-specific strengths (Naive Bayes for recall)
- Consistent weaknesses (KNN across most metrics)

The heatmap's gradient colouring makes patterns immediately apparent, facilitating informed model selection based on business priorities.

8.7 Summary of Findings

The comprehensive evaluation across multiple metrics and visualization techniques converges on several key conclusions:

1. Logistic Regression emerges as the optimal model for this churn prediction task, achieving the best F1-score (0.6092) and providing excellent balance between precision (65.84%) and recall (56.68%). Its interpretability and computational efficiency make it particularly suitable for business deployment.

2. Ensemble methods performed well but not exceptionally, suggesting that the relatively simple feature space and moderate dataset size may not fully leverage their complexity.

Random Forest and XGBoost achieved solid performance but did not substantially outperform the simpler Logistic Regression.

3. The precision-recall trade-off is evident across models, with Naive Bayes achieving 86.63% recall at the cost of only 42.69% precision, while SVM achieved 64.44% precision with 48.93% recall. Business requirements should guide whether optimizing for recall (catching more churners) or precision (reducing false alarms) is prioritized.

4. Feature importance analysis validates business intuition, with tenure, contract type, and charges emerging as primary churn drivers. This alignment between model learning and domain knowledge increases confidence in predictions and facilitates stakeholder acceptance.

5. Model performance metrics (F1-scores of 0.51-0.61) indicate moderate predictive power, reflecting the inherent difficulty of predicting human behaviour from transactional data alone. Customer decisions involve factors beyond those captured in available features, including competitor actions, personal circumstances, and subjective service quality perceptions.

6. All models substantially outperform random guessing, with ROC-AUC scores above 0.75 for most models, validating that meaningful predictive patterns exist in the data.

9. Conclusion

This project successfully developed and evaluated a comprehensive machine learning pipeline for customer churn prediction in the telecommunications domain. Through systematic implementation of data pre-processing, feature engineering, dimensionality reduction, unsupervised clustering, and supervised classification with ten distinct algorithms, the study provides actionable insights into predicting and understanding customer churn. The Logistic Regression model emerged as the optimal predictor, achieving an F1-score of 0.6092 with 80.70% accuracy, 65.84% precision, and 56.68% recall. This performance demonstrates meaningful predictive capability while maintaining model interpretability a critical consideration for business deployment where stakeholder understanding and trust are essential. The model's balanced precision-recall trade-off makes it suitable for practical retention campaign targeting, where both false positives (wasted retention spending) and false negatives (missed churn opportunities) carry costs. Feature importance analysis revealed that customer tenure, contract type, and billing characteristics are primary churn drivers, aligning with business intuition and providing actionable guidance for retention strategies. Customers with short tenure, month-to-month contracts, higher monthly charges, and electronic check payment methods represent the highest-risk segments requiring proactive engagement. The clustering analysis provided complementary insights into customer segmentation, identifying distinct groups with varying churn propensities. These segments can inform targeted marketing strategies, service customization, and resource allocation for customer relationship management. The comparison of ten diverse algorithms revealed interesting patterns: simple linear models (Logistic Regression) performed competitively with complex ensemble methods (Random Forest, XGBoost), suggesting that the feature space, after appropriate engineering and encoding, exhibits primarily linear relationships with churn. This finding has practical implications, as simpler models are easier to deploy, maintain, and explain to non-technical stakeholders. The moderate F1-scores across all models (0.51-0.61) reflect the inherent complexity of predicting human behavior and the limitations of relying solely on transactional and demographic data. Customer churn decisions involve subjective factors service quality

perceptions, competitor offerings, personal circumstances, customer support experiences that may not be fully captured in available features. This suggests that model predictions should complement, rather than replace, human judgment in customer retention strategies. From a business perspective, the developed system offers several deployment pathways. The model can serve as a decision support tool, flagging high-risk customers for manual review by retention specialists. Alternatively, it can automate low-touch interventions, triggering personalized retention offers, targeted communications, or service recommendations for predicted churners. The model's probability outputs enable risk scoring and prioritization, allowing customer service resources to focus on the highest-value, highest-risk customers. The project demonstrates that machine learning can provide meaningful value in customer retention management, transforming reactive churn responses into proactive retention strategies. By identifying at-risk customers before they churn, telecommunications companies can implement timely interventions that improve customer satisfaction, reduce churn rates, and enhance profitability.

10. Limitations

While the project achieved its objectives, several limitations should be acknowledged:

Dataset Size and Diversity: With 7,043 samples, the dataset is moderately sized but may not fully represent the diversity of a large telecommunications company's customer base. Demographic diversity, geographic variations, and service offering differences in real-world operations may not be adequately captured. Larger datasets with greater variety could enable more robust model training and better generalization.

Feature Limitations: The dataset lacks several potentially important predictors of churn:

- **Service quality metrics:** Call drop rates, connection speeds, outage history
- **Customer support interactions:** Number and outcomes of support tickets
- **Competitive information:** Availability of competitor services in customer's area
- **Customer sentiment:** Satisfaction scores, survey responses
- **Temporal patterns:** Seasonal variations, service usage trends over time

Including such features could substantially improve prediction accuracy.

Temporal Considerations: The dataset represents a snapshot in time without temporal sequencing. Churn prediction could benefit from time-series modelling approaches that capture evolving customer behaviour patterns, such as declining usage, increasing support contacts, or payment delays preceding churn.

Model Complexity vs. Dataset Size: The moderate dataset size may have limited the effectiveness of complex models like Neural Networks and ensemble methods, which typically require larger training sets to demonstrate their full potential. The strong performance of Logistic Regression may partially reflect this dataset-size constraint rather than indicating that simpler models are universally optimal for churn prediction.

Class Imbalance: While the 73:27 non-churn:churn ratio is not severely imbalanced, it still influenced model behaviour. More sophisticated imbalance-handling techniques beyond stratified sampling such as SMOTE, cost-sensitive learning, or ensemble balancing might improve minority class detection.

Hyperparameter Tuning: Models were trained with default or basic hyperparameters rather than extensive tuning through grid search or Bayesian optimization. Systematic hyperparameter tuning could improve performance for all models, particularly complex ones like XGBoost, Random Forest, and Neural Networks.

Deployment Considerations: The study focused on offline model development and evaluation rather than deployment aspects such as:

- Real-time prediction latency requirements
- Model monitoring and drift detection
- A/B testing for retention campaign effectiveness
- Integration with customer relationship management systems
- Ethical considerations regarding automated decision-making

Generalization Across Contexts: The model was trained on a specific telecommunications dataset and may not generalize to other telecom companies with different service portfolios, customer demographics, or market conditions. Similarly, the insights may not transfer to other industries with different churn dynamics.

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